

User-Elicited Gestural Interaction With Future In-Car Holographic 3D Displays

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ABSTRACT

Holographic 3D displays (H3D) have the potential to expand the interaction space for in-car infotainment systems by providing a larger depth range than other state of the art 3D display technologies. This work explored how non-expert users would interact with non-driving related tasks tailored to H3D visualization. In a gesture-elicitation study, $N = 20$ participants proposed mid-air gestures for a set of 33 tasks (referents) displayed either within or outside of participants' reach. In a follow-up reverse-matching study with $N = 21$ participants, the resulting set of most mentioned gestures was refined. The final gesture set shows that techniques elicited for other 3D technologies are applicable to interaction with future in-car H3D displays.

CCS CONCEPTS

• **Human-centered computing** → **Gestural input**; *Mixed / augmented reality*; Interaction design process and methods.

KEYWORDS

datasets, neural networks, gaze detection, text tagging

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1 INTRODUCTION AND RELATED WORK

Holographic 3D displays (H3D) provide the largest depth without loss of resolution or conflicting depth cues associated with other state of the art 3D display technologies [10]. Level 4 automation could allow passengers to engage in non-driving related tasks [2, 16] which could support the inclusion of more complex gestures with greater range of motion. Gestures will usually match the application they were primarily designed for [1, 6]. It is therefore useful to evaluate, whether existing interaction techniques can be re-purposed for new emerging applications [1]. The gesture elicitation method [6, 17, 18] is a suitable user-centered approach to find intuitive interaction techniques. The aim of the presented work was to investigate how non-expert users would use gestures to interact with

holograms appearing within and outside of reach. Because the H3D technology is still in development, a Microsoft Hololens 1 was used as a display device.

2 STUDY 1: GESTURE ELICITATION

20 participants (13 m) with a mean age of 39.65 ($SD = 11.93$) years without background in interaction or user interface design were shown holograms in “before” and “after” states representing the effects of 33 referents for infotainment control, manipulation of maps and manipulation of 3D objects in 3D space. Participants were instructed to think aloud and produce at least two suitable gestures for each presented referent, then choose their preferred option [15] and rate its suitability and ease of use, each on a 7-point Likert scale ($-3 =$ strongly disagree, $0 =$ undecided, $3 =$ strongly agree) [4, 9, 18]. The general study set-up is presented in Figure 1.

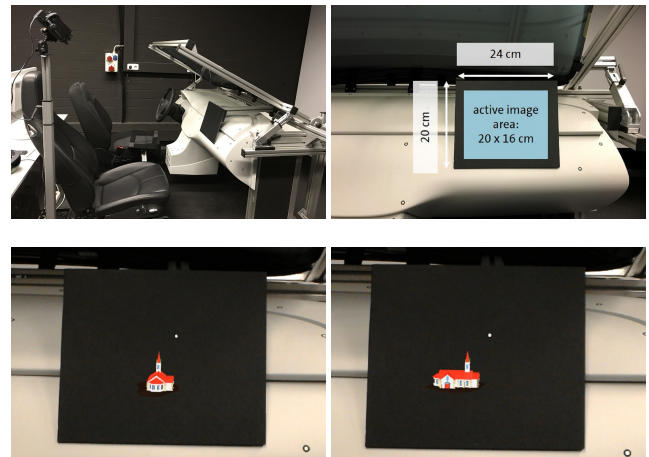


Figure 1: Top row: Holograms were displayed within a 20 x 16 cm virtual display screen on a black placeholder plate attached to the dashboard of a seating buck 80 cm in front of the seated participant. Bottom row: Example hologram of a 3D object in its ‘before’ (left image) and ‘after’ (right image) state representing the effect of the referent ‘rotate around vertical axis’.

In the within reach condition, holograms appeared up to 40 cm in front of the placeholder and up to 120 cm behind the placeholder in the outside of reach condition. Each participant completed both conditions in a randomized order (within-design).

160 gesture groups in the within reach and 223 groups in the outside of reach condition were identified following the loosened

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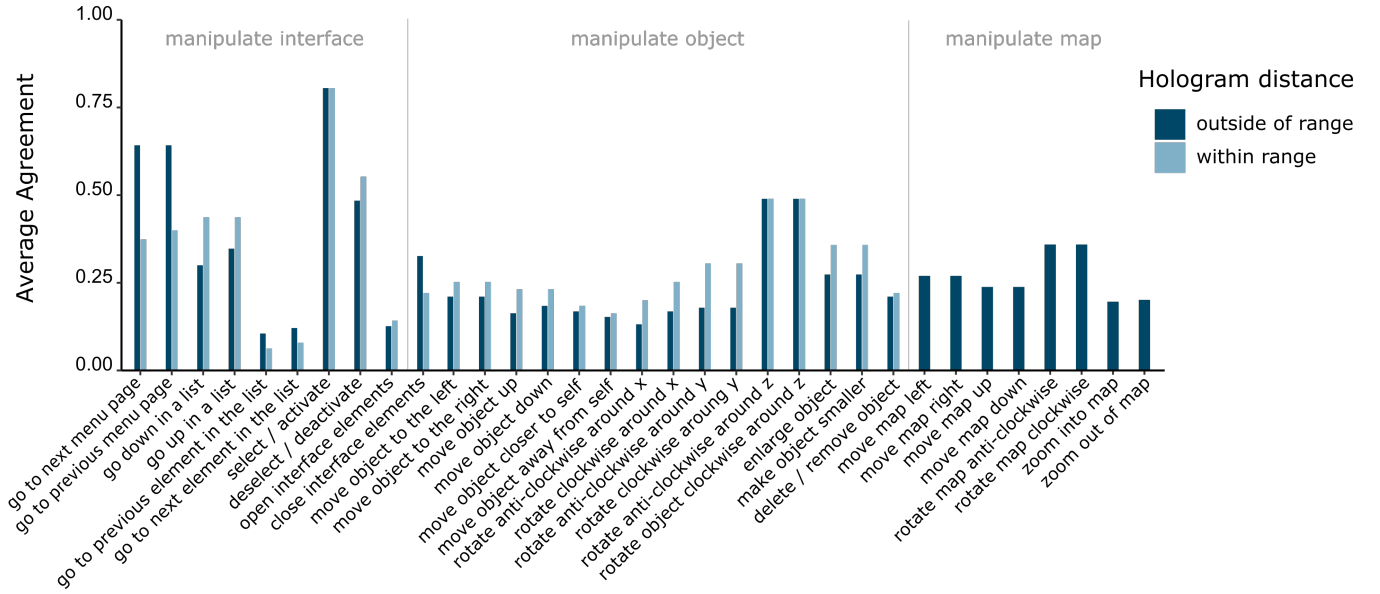


Figure 2: Average agreement rates for each referent in each condition. Map related functions were presented only in the outside of range condition due to the scene size. The axis were defined according to the left-handed coordinate system. The referent ‘deselect/deactivate’ was later removed from the referent list based on participants’ feedback.

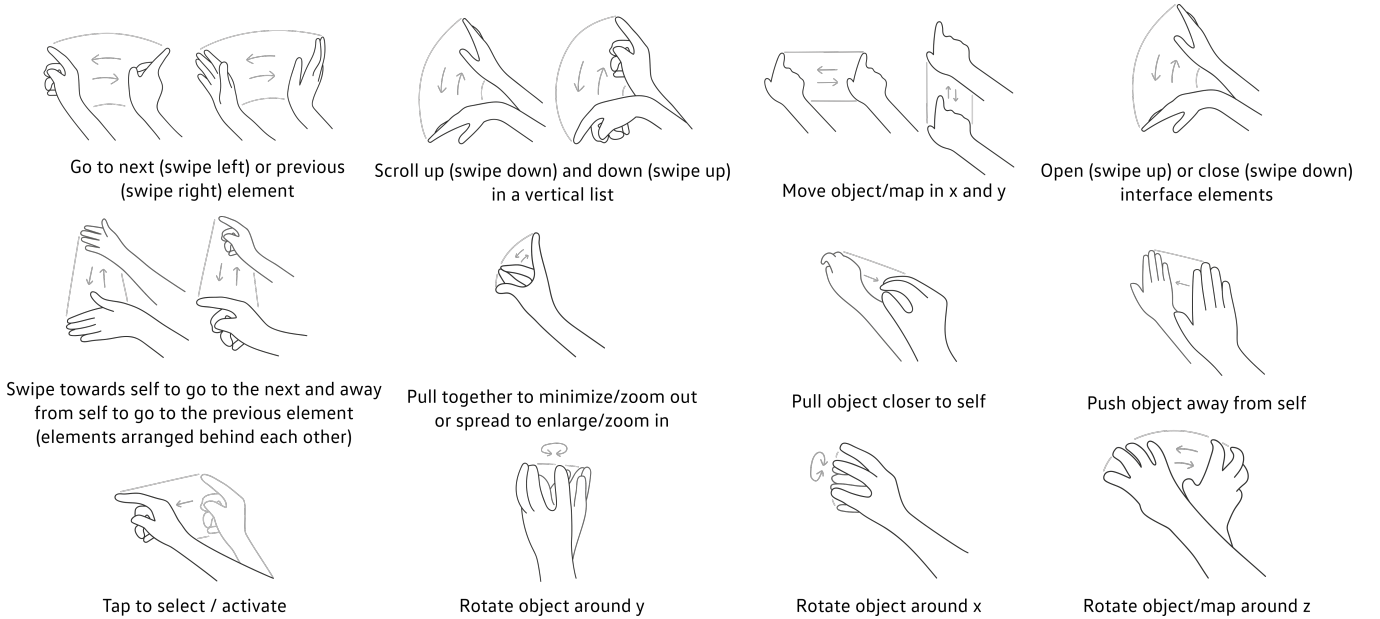


Figure 3: Refined user-elicited gesture set. The axes were defined according to the left-handed coordinate system.

constraint [9] which allows gestures to be similar but not necessarily identical within each group. The overall average agreement rate (AR) [13] and chance-corrected agreement (K_F) [12] was $AR = 0.312$ ($SD = 0.168$, $K_F = 0.295$) in the within and $AR = 0.288$ ($SD = 0.168$, $K_F = 0.258$) in the outside of reach condition, which represents a medium agreement [13]. Agreement rates per referent in each condition are illustrated in Figure 2. Most frequently proposed

gesture groups were included in the gesture set, resulting in 29 gesture-referent pairings. 9 referents produced different gestures for within and outside of reach interaction.

3 STUDY 2: REVERSE-MATCHING

21 participants (17 m) with a mean age of 38.76 ($SD = 10.46$) years without background in interaction or user interface design viewed

videos of each gesture from the elicited gesture set and chose the referent for which that gesture seemed most suitable [6]. Suitability of a gesture was measured by calculating the number of correct pairings between the gesture and its corresponding referent [4]. Gesture-referent pairings with suitability scores below 10 were removed from the gesture set. Swipe motions with hand or finger were matched to the same referents and were hence grouped together, resulting in a final set of 23 gesture-referent pairings (Figure 3).

4 DISCUSSION

A majority of elicited gestures resembled touchscreen techniques or physical manipulations of real-world objects, indicating a strong influence of previous interface experience [4, 7, 9, 11, 14]. The final gesture set reveals similarities to gestures elicited for 3D interaction with mixed reality applications [8, 9] and in-car stereoscopic 3D displays [15]. This confirms that many existing 3D techniques can be applied to in-car interaction with H3D, indicating an opportunity to create holistic design frameworks across multiple systems. The general lack of touch feedback remains a major critique point during mid-air interaction. Future studies of H3D interaction could aim to investigate the benefit of various feedback modalities, as well as evaluate the efficiency of proposed interaction techniques in a moving vehicle. Cultural background was not accounted for in this study, but could have influenced gesture proposals and preferences [3, 5].

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